TECHNICAL REPORT

F1 Model Simulator

Closed-loop lap simulation of a 2026-regulation Formula 1 car on Silverstone, in MATLAB/Simulink

Model version	v7
Environment	MATLAB / Simulink R2026a, Vehicle Dynamics Blockset
Result	Verified closed-loop test bench: 8/8 regression checks, 2.0 m maximum line error, 45 parameters with declared status
Date	November 2026
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Repository	github.com/Julian9911

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1. Summary

This report documents a lap-time simulator of a 2026-regulation Formula 1 car. A virtual pilot drives the car around Silverstone in a closed loop, and the result is compared with public telemetry of a real qualifying lap. The simulator is a platform that can be extended: the model is deliberately simple at the vehicle level (rigid body, lumped aerodynamics), but every parameter has a declared source or status, every change was tested on its own, and the limitations are stated.

8 / 8	2.0 m	45 / 45	316 km/h
regression checks pass	max lateral error to the line	parameters with a declared status	top speed (real 317)

What the model does well. It follows the planned line with a maximum error of 2.0 m, reaches the real top speed once the active aerodynamics are modelled, and reproduces the corner and straight behaviour with a limited number of well-documented assumptions. **What it does not do.** It does not close the last 6.8 s: the gap is concentrated in medium and fast corners and is only partly explained (section 7). The ERS, wet track, tire compounds and wear are out of scope by decision (section 8).

2. Objectives and scope

The goals, in order of priority:

1. A model whose behaviour can be explained: each effect that changes the lap time has a physical reason and a test.
2. Validation against real data, with honest reporting of what does not match.
3. A platform: clear subsystems, a regression test and a parameter register so new effects (ERS, tires, wet track) can be added without breaking what exists.

Out of scope: matching the real lap time at any cost, using team data, race simulation (consumption, degradation, strategy) and 3D visualization. The data used is public: the 2026 technical regulations and commentary, secondary aerodynamic sources, OpenStreetMap geometry of the circuit and public telemetry of a real lap.

3. System architecture

The model is a closed loop with three subsystems: a **Controller** (the pilot), a **Plant** (the car) and an **Instrumentation** block (lap timer). The data the model needs (reference line, speed profile, finish line, vehicle parameters) is created in the workspace by a single entry-point script.

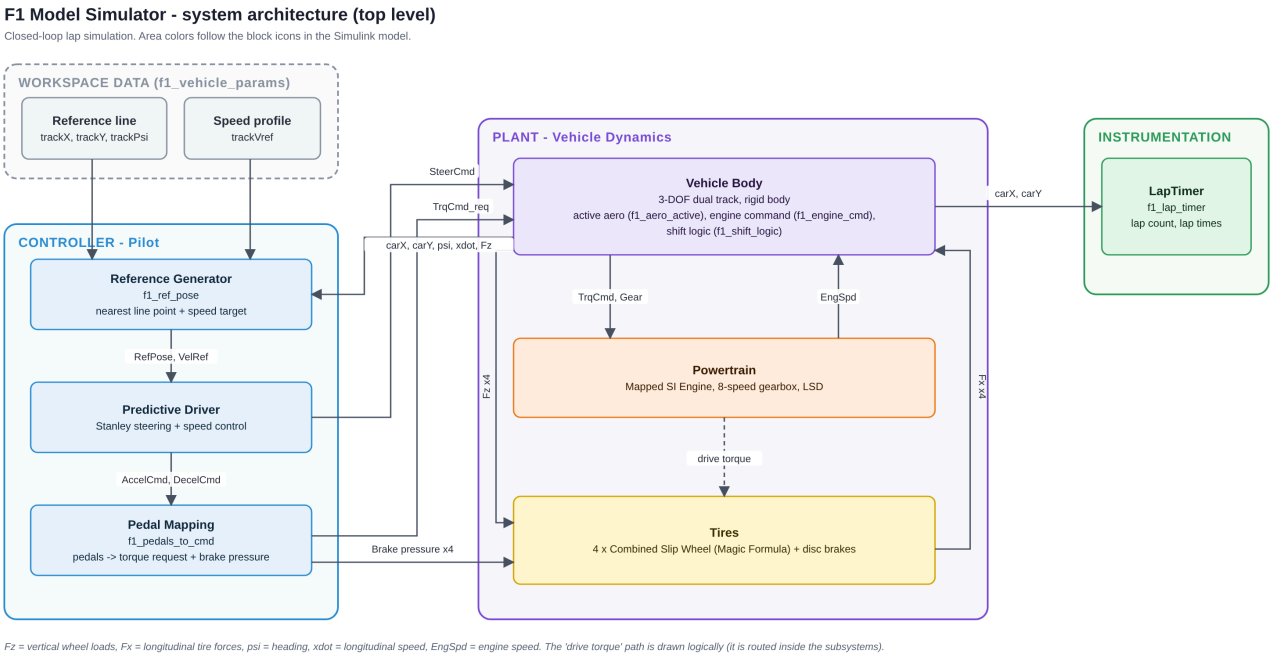


Figure 1 - Top-level architecture. Colors follow the icons of the Simulink blocks.

The plant combines a 3-DOF vehicle body, a power train (engine map, gearbox, limited-slip differential) and four wheels. The engine torque request is limited by the grip available at the rear axle and by the rev limiter; braking is limited per wheel. Both limits use a friction circle so that cornering grip is not spent twice.

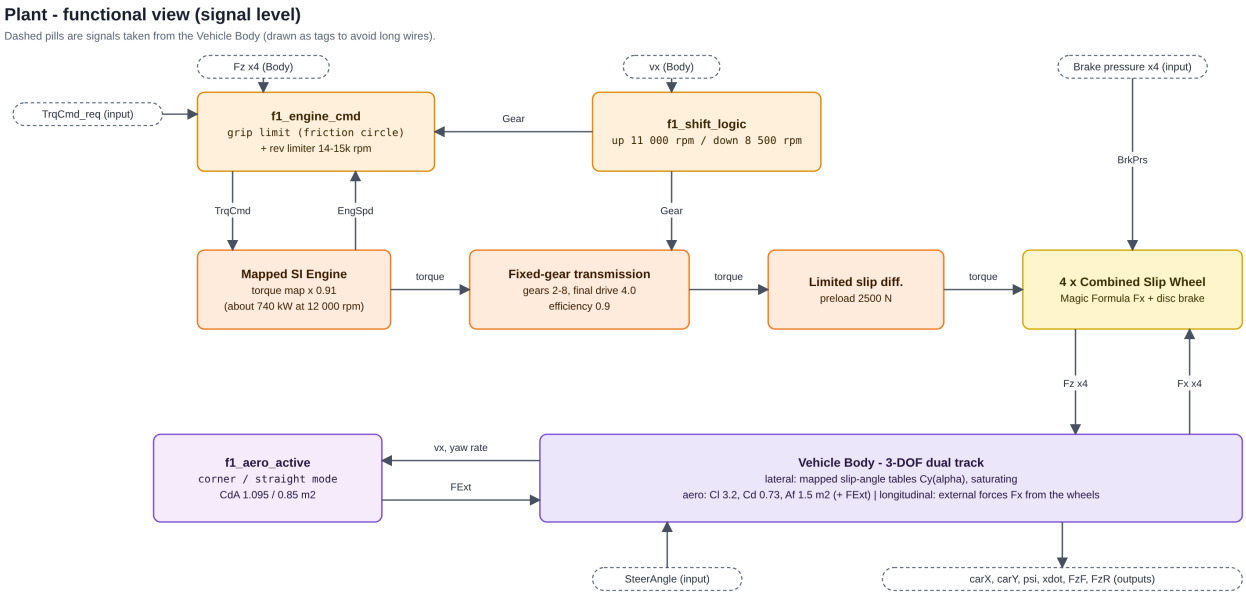


Figure 2 - Plant at signal level. Dashed pills are signals taken from the Vehicle Body.

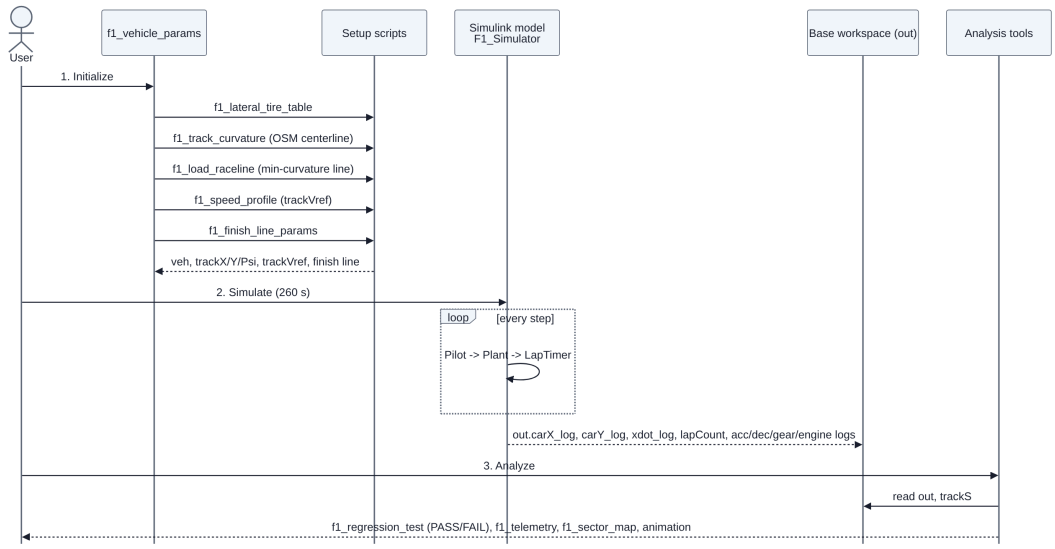


Figure 3 - Workflow: initialize, simulate, analyze.

The model as built in Simulink. The top level groups the *Analysis tools* buttons (telemetry, sector map, animation), the Controller, the Plant and the Instrumentation; the blocks carry the same colors and icons as the diagrams above.

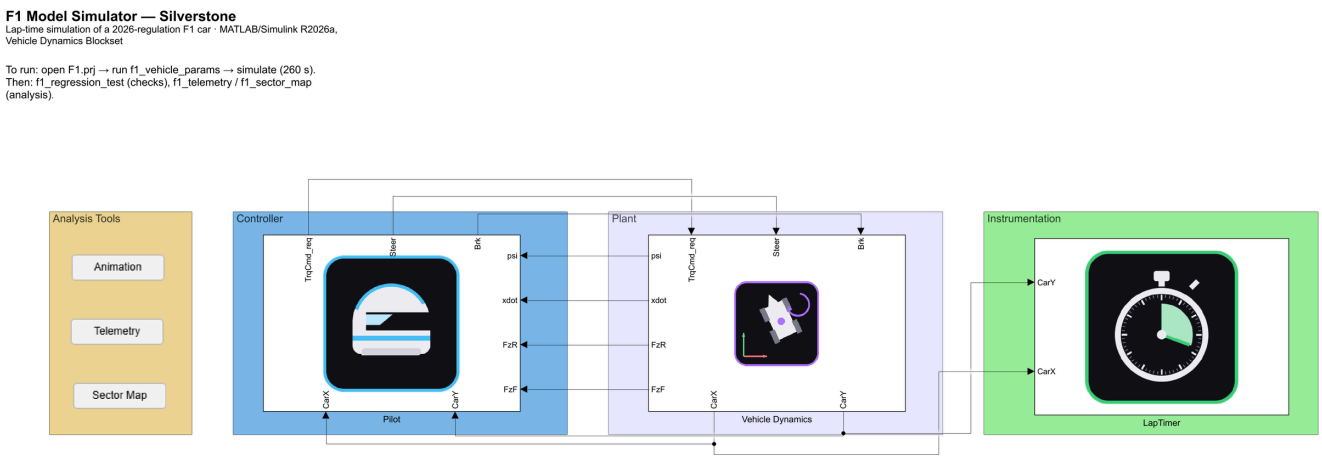


Figure 4 - Simulink model, top level.

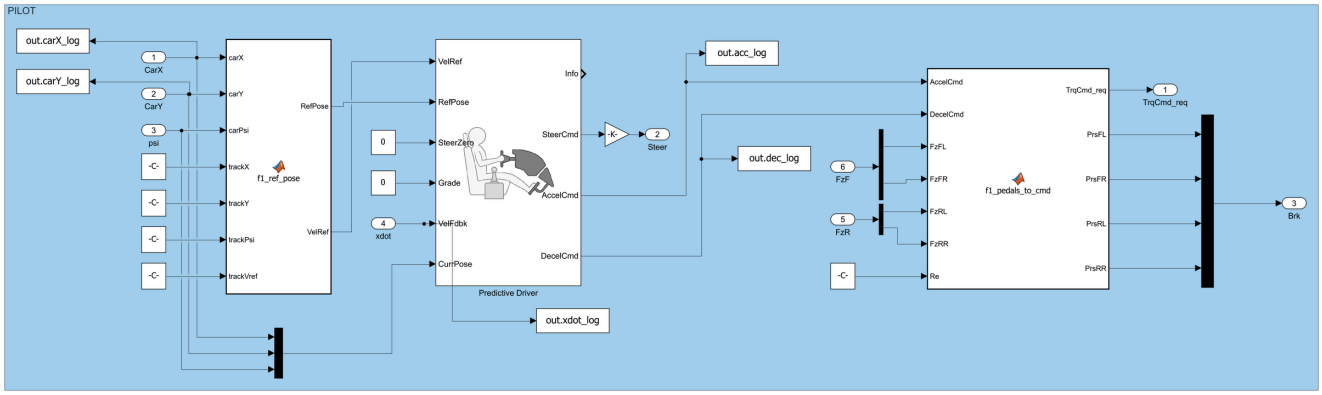


Figure 5 - Controller (Pilot): reference pose and speed target, Predictive Driver, pedal-to-command conversion.

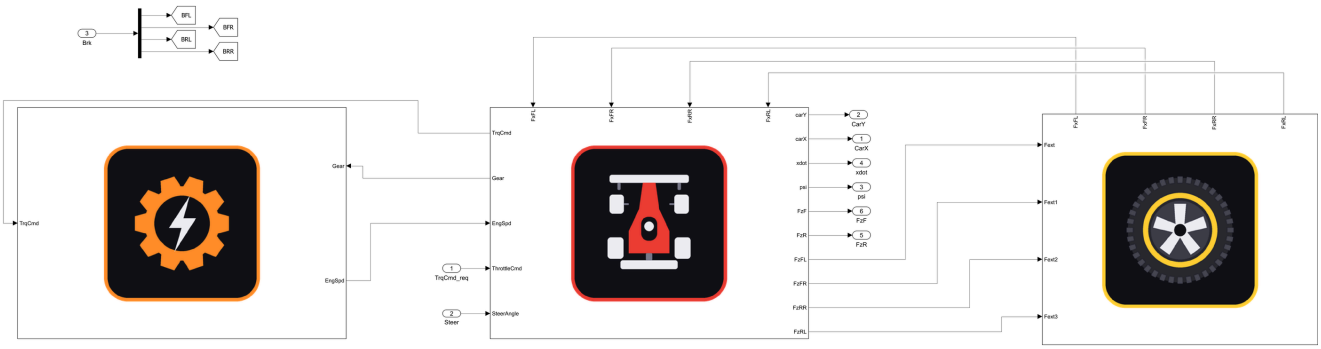


Figure 6 - Plant (Vehicle Dynamics): power train, vehicle body with engine command and shift logic, wheels.

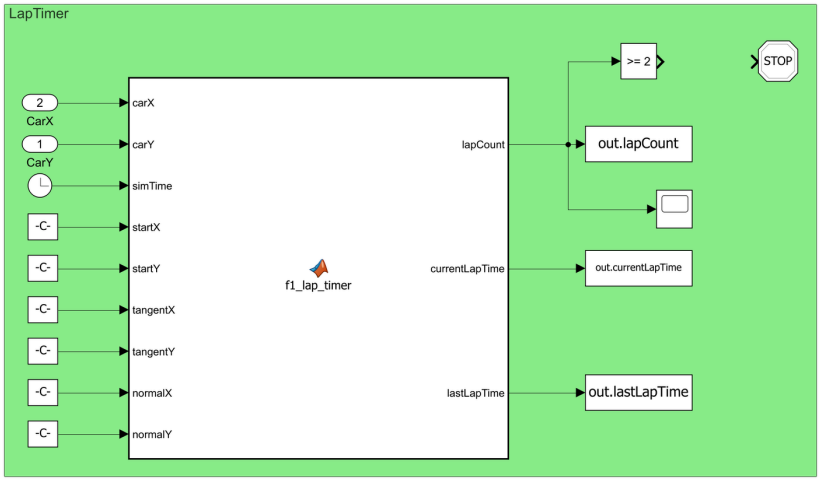


Figure 7 - Instrumentation (LapTimer): finish-line detection, lap counter and a stop after two laps.

4. Models

4.1 Vehicle body

The Vehicle Body 3DOF Dual Track block gives planar motion (longitudinal, lateral and yaw) with an individual vertical load on each wheel. Load transfer comes from accelerations through the load-transfer height *h* and the track width. There is no suspension, roll, pitch or ride height. Mass (770 kg) and wheelbase (3.4 m) are the 2026 regulation values; the weight distribution (55 % rear), CG height, track width and yaw inertia are estimates.

The block is used in the "external longitudinal forces" mode: the wheels compute the longitudinal force (Magic Formula) and the body takes it as an input, while the lateral force comes from the body's own slip-angle table.

4.2 Tires

The lateral force per wheel comes from a tabulated cornering stiffness Cy(α), designed so that the force saturates like a Magic Formula curve:

> $F_y = C_y(\alpha) \cdot \alpha \cdot \mu \cdot F_z \,/\, (2\, F_{z,nom}), \quad \mu_y(\alpha) = D \sin(\, C \operatorname{atan}(\, B\alpha - E\,(B\alpha - \operatorname{atan} B\alpha)\,)\,)$

The peak is 1.7 on the front axle and 1.8 on the rear, so the front saturates first (stable understeer at the limit). The initial slope is 20 and 24 per radian, in the 20-25 range of racing tires. The block scales this force linearly with load, so grip does not drop with load as a real tire does (limitation L-3). Longitudinal force uses the Magic Formula of the wheel block (peak 1.7, `FZMAX` 15 000 N).

4.3 Aerodynamics, including the active modes

Aerodynamics are lumped: *downforce* = ½ ρ ClA v^2 with ClA = 3.2 × 1.5 = 4.8 m^2, and *drag* = ½ ρ CdA v^2 with Cd 0.73 (CdA 1.095 m^2) in corner mode. ClA 4.8 m^2 lies inside the 3.2-6.4 m^2 range of public secondary sources.

The 2026 car has a corner mode (high downforce) and a straight mode (low drag). Cd and Cl of the body block are fixed during a simulation, so the block represents the corner mode and the straight mode is added through the external force port:

> $dF_x = \tfrac{1}{2}\,\rho\, v^2\,(\,CdA_{corner} - CdA_{straight}\,), \quad CdA_{corner} = 1.095\ m^2,\ CdA_{straight} = 0.85\ m^2$

The straight mode is active when the lateral acceleration (approximated by $|r\,v_x|$) is below 3 m/s² and speed is above 40 m/s, and it returns to corner mode above 6 m/s² (hysteresis). $CdA_{straight}$ is a **calibration** to the real 90th-percentile acceleration at 70-90 m/s (range 0.8-0.9), not a published value. Only drag is changed; the reduction of downforce on the straights is not modelled.

4.4 Power train and controls

Engine. A mapped engine whose torque is scaled by 0.91 gives about 740 kW at 12 000 rpm, against the 750 kW total of the regulations (about 400 kW combustion plus 350 kW electric). The map stands for the whole electrified power train, without energy limits. In practice the engine does not exceed about 11 700 rpm (shift logic and top gear), where the map delivers about 700 kW; the map reaches about 900 kW only above 14 000 rpm, which the car never uses.

Gearbox. Gears 2 to 8 (first gear is unused in a flying lap) with ratios derived from the median rpm/speed of each gear in the real telemetry; a final drive of 4.0 and an efficiency of 0.9. Shifts happen at 11 000 rpm (up) and 8 500 rpm (down); the real up-shifts have a median of about 10 900 rpm.

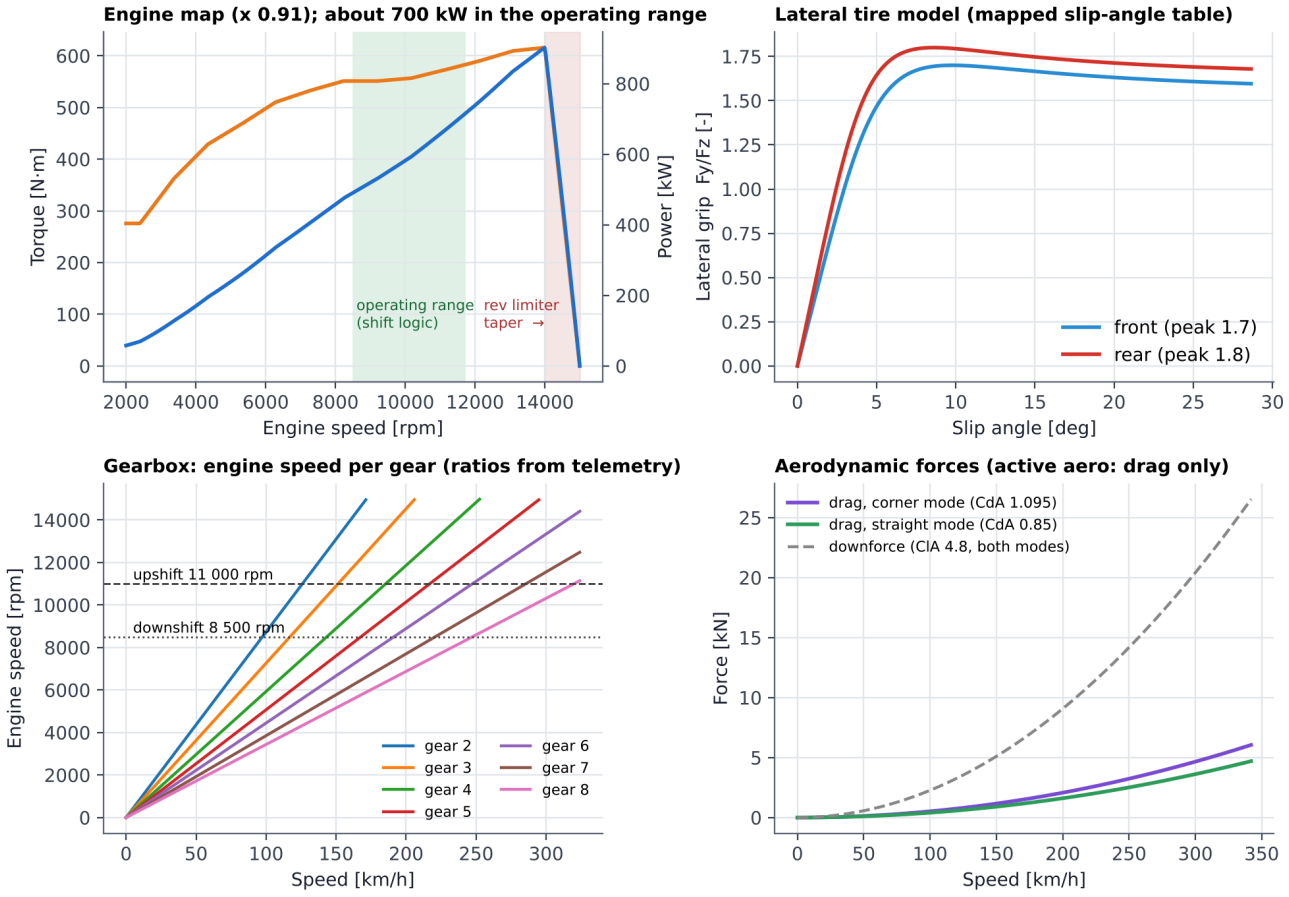


Figure 8 - Model characteristics reproduced from the constants in the MATLAB files: engine map, lateral tire model, gearbox, aerodynamic forces.

Differential. A limited-slip differential with a 2500 N clutch preload (an assumption). The lap time did not change measurably between 500 and 12 500 N (95.18 / 95.16 / 95.24 s).

Traction and brake limits (friction circle). With an open or lightly locked differential the axle can only push what the weaker rear wheel holds. The available grip is what is left after the lateral grip that the corner already uses:

$$\mu_x = \min(\,\mu,\ \sqrt{(\mu_{pk}^2 - \mu_y^2)}\,), \quad \mu_y = |\Sigma F_{z,right} - \Sigma F_{z,left}| \cdot track / (2\,h\,\Sigma F_z)$$

μ_y is estimated from the lateral load transfer. Before this change, higher grip made a rear wheel spin on corner exit and another lock under braking, because the limiters assumed all the grip was available. With `useFC = false` the previous version is reproduced exactly (checked).

4.5 Pilot and reference line

A Predictive Driver (Stanley lateral control, look-ahead 40 m) follows a reference pose and a reference speed. `f1_ref_pose` returns the closest point of the line (searching only ±100 m around the last one, so that the pilot cannot lock onto another part of the circuit that passes nearby) and its speed target. The reference line is a minimum-curvature line computed inside ±4 m of the OpenStreetMap centerline; the real GPS path is deliberately not used (decision D-10).

4.6 Speed profile

The speed target is computed offline for each 2 m of the line in three steps. First, the corner limit including downforce:

> $v_{max} = \sqrt{(\mu\, g / (\kappa - \mu\, k))}, \quad k = \tfrac{1}{2} \rho\, ClA / m, \quad \mu = 1.7$

Where $\kappa < \mu\, k$ (radius above about 154 m) downforce sustains any speed and only the speed cap applies; this makes the profile very sensitive to the radius in fast corners. Second, a backward pass limits speed to what can be braked at 20 m/s². Third, a forward pass uses an acceleration table **computed** from the engine map, gearbox, drag, rolling resistance and traction limit. Both passes run twice around the circuit so that the lap closes. The computed table agrees with the acceleration measured in the simulator within 0.5 m/s² above 40 m/s and is optimistic by 1.0-1.5 m/s² below 35 m/s.

4.7 Lap timing

The lap timer detects crossings of the finish line with a 30 m lateral tolerance and a 2 s grace period. The second lap, which starts at speed, is the one evaluated.

4.8 Two layers and the parameter source (v7)

The project has two layers. Layer 1, `f1_speed_profile`, is a quasi-steady-state point-mass lap-time simulation that proposes a speed profile (about 91.2 s, the ideal lap). Layer 2 is the Simulink closed-loop vehicle that follows it (94.93 s). The 3.7 s between them is the tracking gap. The model is a MIL test bench: the driver follows the profile, and the grip level is chosen by hand because an automatic limit search would automate one parameter out of many (decision D-15).

All physical values are defined once in `f1_vehicle_params.m` (structs `veh`, `aero`, `tire`, `pwr`, `brk`). Block dialogs use workspace expressions and the MATLAB Function blocks receive flat aliases `fp_*` as Parameter data (D-14). The regression test stayed at 8/8 and 94.93 s after each step of the refactor.

4.9 Sensitivity of the lap time (v7)

Each parameter was increased by 5 % with the full closed loop (`f1_sensitivity_sweep`). The profile-only column shows the effect on the ideal lap.

Parameter (+5 %)	Closed loop	Profile only
Cornering grip mu	−1.76 s	−2.05 s
Engine torque scale	−0.97 s	−0.51 s
Mass	+0.96 s	+1.24 s
Downforce ClA	−0.77 s	−0.90 s
Drag CdA	+0.06 s	+0.21 s

Grip dominates. The CdA effect in the closed loop is below the resolution of the model and should not be interpreted.

4.10 Internal states (v7)

`f1_internal_states` shows per-wheel vertical load, longitudinal and lateral force, a force utilization index and the g-g diagram, in a full view and a simplified view, to separate what cannot be measured on a real car from what can. Peaks are about 4.5 g lateral, 4.2 g braking and 1.8 g traction. Caveats: the signals come from the Vehicle Body Info bus, not from the Combined Slip outputs; utilization is defined as |F| / (mu_pk Fz) by the script; there is no trail braking (the transition from braking to cornering is abrupt); the nature of the real AccX/AccY channels (measured or derived) is unconfirmed.

4.11 Corner map (v7)

`f1_corner_map` draws the circuit from above with the line the model drove, colored by braking, throttle and lift. The corners are detected as deep speed minima on the simulated lap (treating the lap as circular); for each one the tool gives the minimum speed, the peak lateral g, the braking zone, the speed after the apex and the maximum distance to the reference line, and plots speed, pedals and accelerations around it against the real reference lap. A slider moves a point along the circuit. Track width is illustrative (OpenStreetMap has none), corners are numbered rather than named, and the real lap is compared by position along the lap.

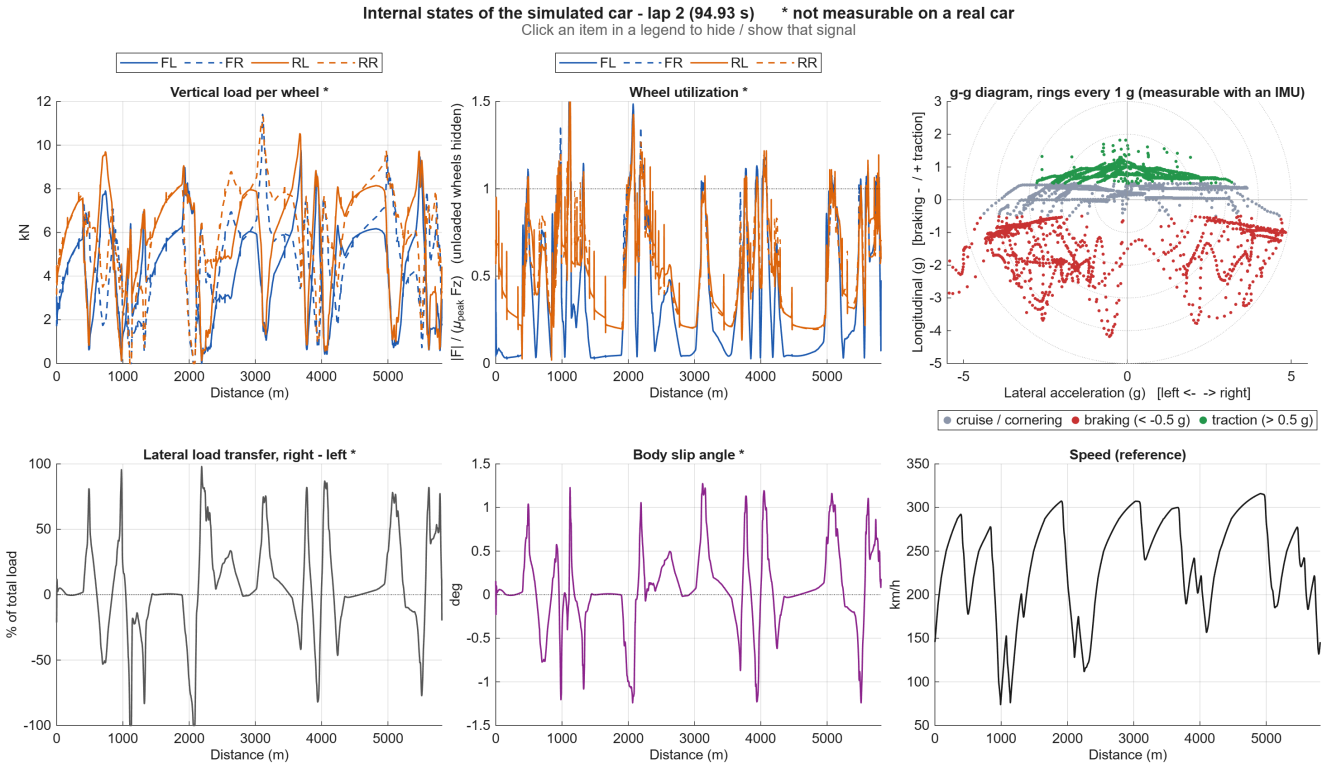


Figure 9 - Internal states of the simulated car, lap 2 (* marks signals not measurable on a real car).

5. Parameters and their provenance

The model has 45 documented parameters (see the parameter register). They are classified by how much trust they deserve:

Status	Meaning	Count
REGULATION	Taken from the 2026 technical regulations (mass, wheelbase)	2
SOURCED	Value or range from a cited public source	3
TELEMETRY	Derived from the public reference telemetry (gear ratios, shift points)	2
CALIBRATED	Tuned against a reference measurement (engine scale, straight-mode drag, braking level)	3
DERIVED / BLOCK / DESIGN	Computed, block settings and algorithm choices	15
ESTIMATED	Engineering judgement without a source (tire peaks, CG height, track width, LSD preload, brake system)	20

The honest reading is that only a handful of parameters come from regulations or data; most tire and chassis values are estimates. The parameters that most influence the lap time (tire peaks, CIA, engine scale) were chosen inside published ranges and tested one at a time. One open item is listed in the register: several values are repeated by hand in more than one place (OI-1). Note that the CG height, 0.35 m, is higher than the 0.25-0.30 m usual in the literature; it is kept because every result was obtained with it.

6. Verification

Verification asks whether the model does what it is meant to do. It is automated in `f1_regression_test.m`, which runs after every change and prints PASS or FAIL per criterion. Each criterion is traced to a requirement (see the requirements document).

Criterion	Threshold	Result (v7)
Complete laps	at least 2 in 260 s	pass
Lap time	94.9 ± 1.5 s	94.93 s
Lateral error to the reference line	max < 3.0 m, 95th percentile < 1.8 m	2.01 m / 1.54 m
Distance travelled	circuit length ± 3 %	pass
Signals finite	no NaN or Inf	pass
Physical plausibility	top speed < 342 km/h; no stops	316 km/h
Repeatability	consecutive laps within 1.5 s (3+ laps)	not evaluated with 2 laps

Other verification evidence: the friction circle switch reproduces the previous version exactly when disabled; the connection of the external force port was checked by setting the force to zero, which reproduced the previous lap time (95.28 s) before the straight mode was switched on; and the rear-wheel slip stays within about −0.09 to +0.08 outside of short peaks (up to 0.18, twice per lap), with no wheel spin or lock-up.

7. Validation against the real lap

Validation asks whether the model resembles the real car. The reference is the fastest qualifying lap of the 2026 British Grand Prix from public telemetry (88.09 s), aligned to the circuit by position (mean distance to the line 2.3 m). An earlier comparison by integrated distance was offset by 85-165 m and was corrected.

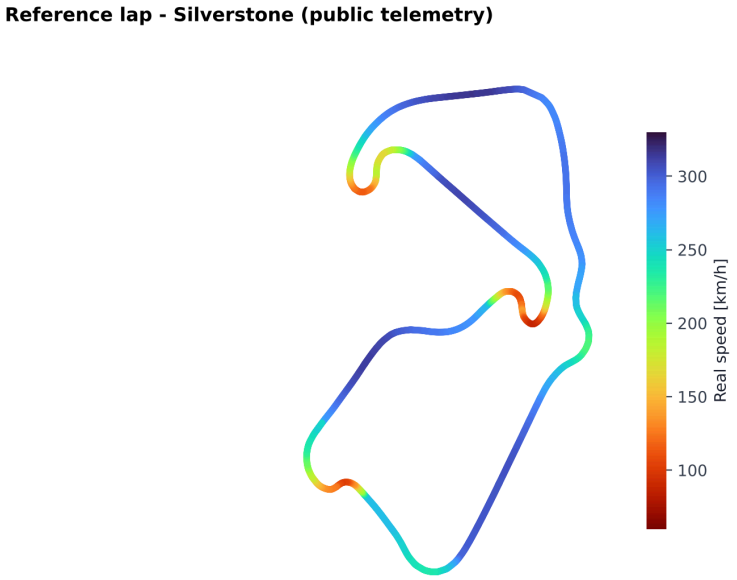


Figure 10 - Reference lap on the circuit, colored by real speed (public telemetry).

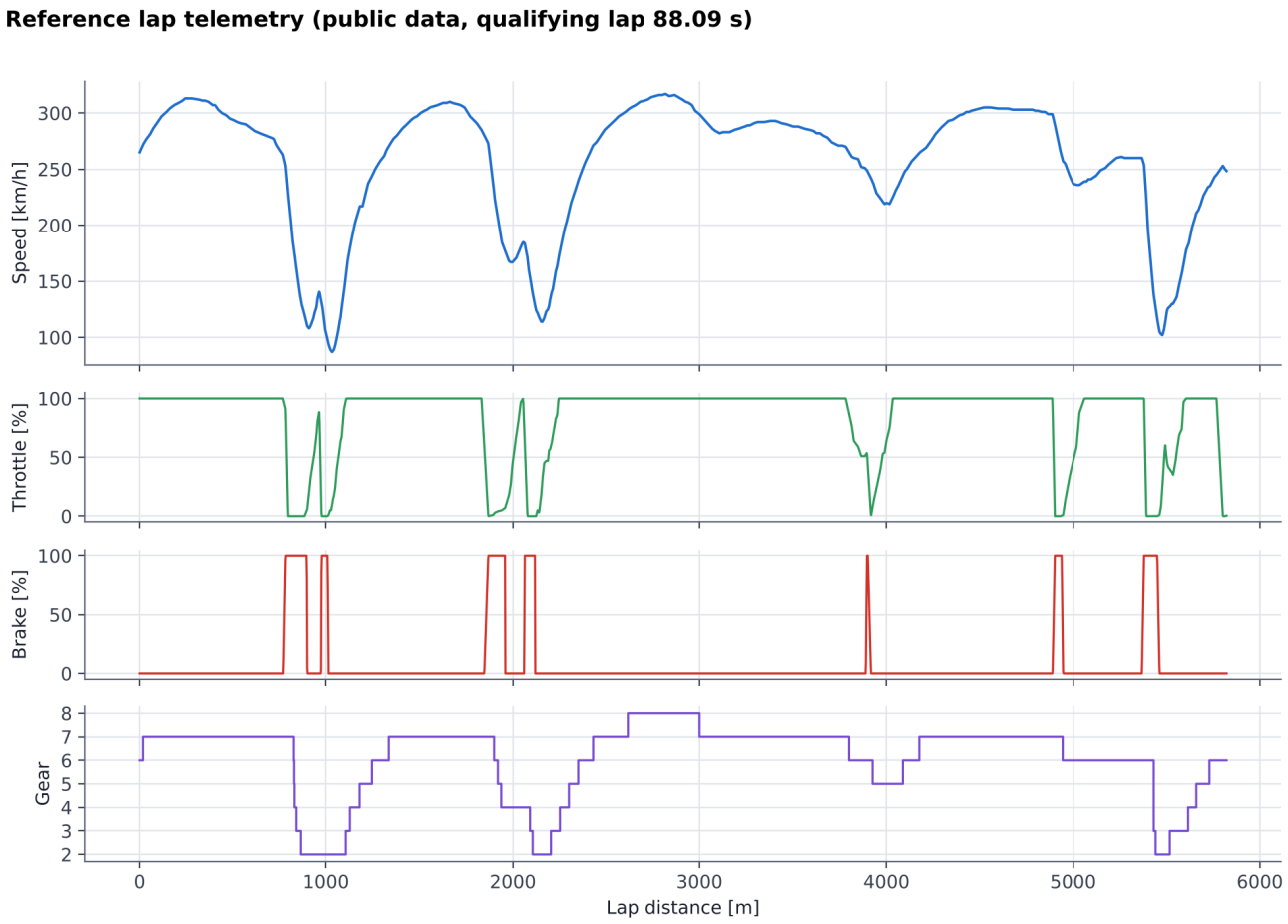


Figure 11 - Reference lap telemetry: speed, throttle, brake and gear against lap distance.

7.1 Result by sector

Sector limits (1660 m and 4420 m along the centerline) are approximate, taken from an image of the sectors. Real sector times are estimates from the aligned telemetry, not official timing.

Sector	Sim	Real	Difference	Top speed sim / real	Min speed sim / real
1 Abbey - Wellington	30.96 s	26.58 s	+4.38 s	292 / 313 km/h	74 / 87 km/h
2 Brooklands - Chapel	43.09 s	39.17 s	+3.93 s	307 / 317 km/h	112 / 114 km/h
3 Hangar - Club	20.87 s	22.34 s	−1.46 s	316 / 305 km/h	132 / 102 km/h

By corner speed class (using the real speed), the simulator is 1.0 s faster in slow sections, 3.5 s slower in medium ones and 4.3 s slower in fast ones.



Figure 12 - Sector map: simulator against the real lap, with time, speed and G per sector.

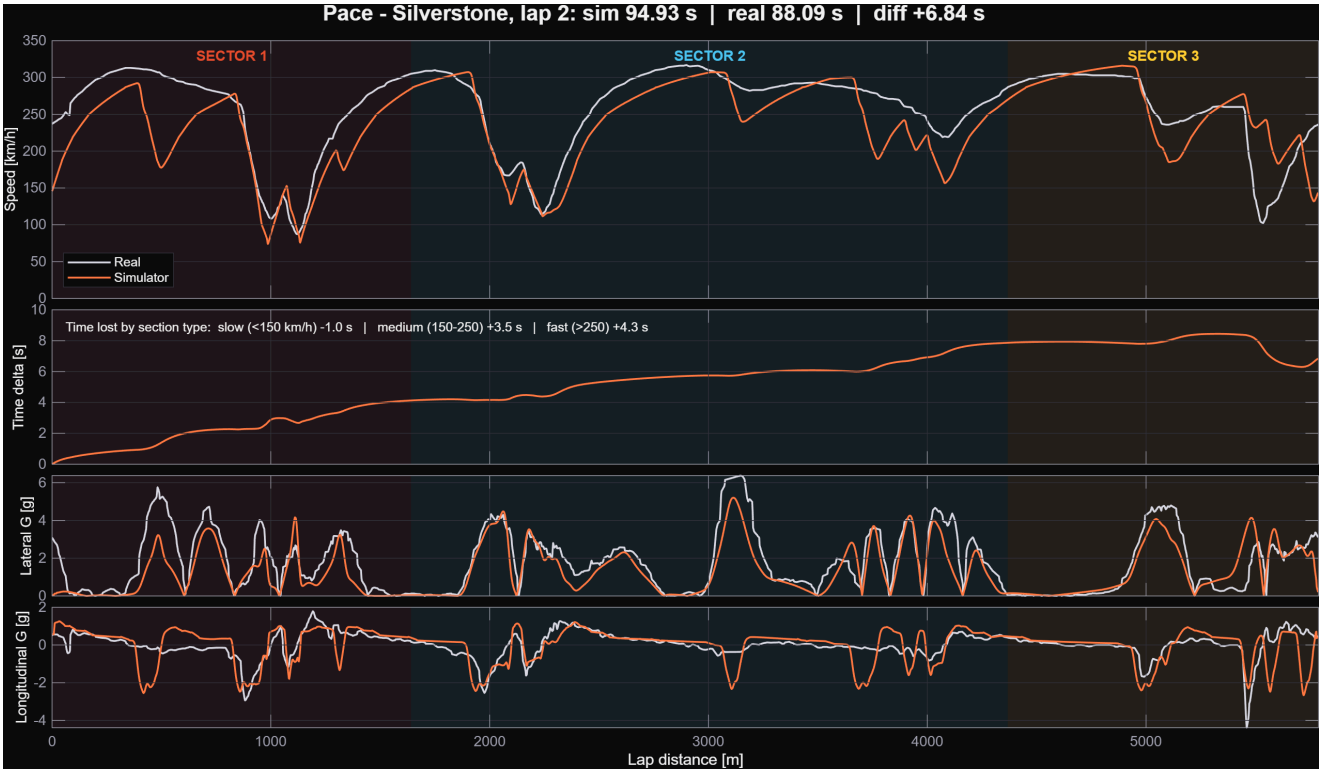


Figure 13 - Pace: speed, accumulated time difference, lateral and longitudinal G against lap distance.

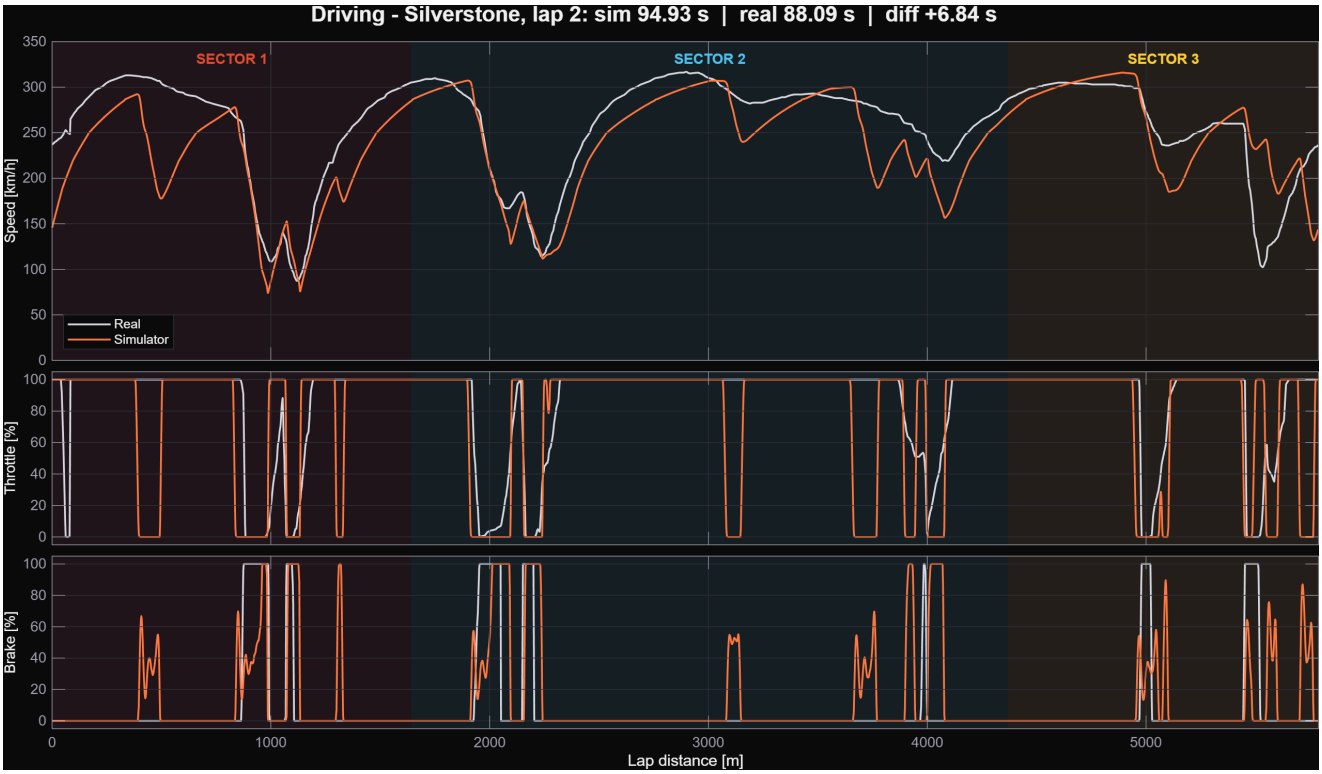


Figure 14 - Driving: speed, throttle and brake, simulator against real.

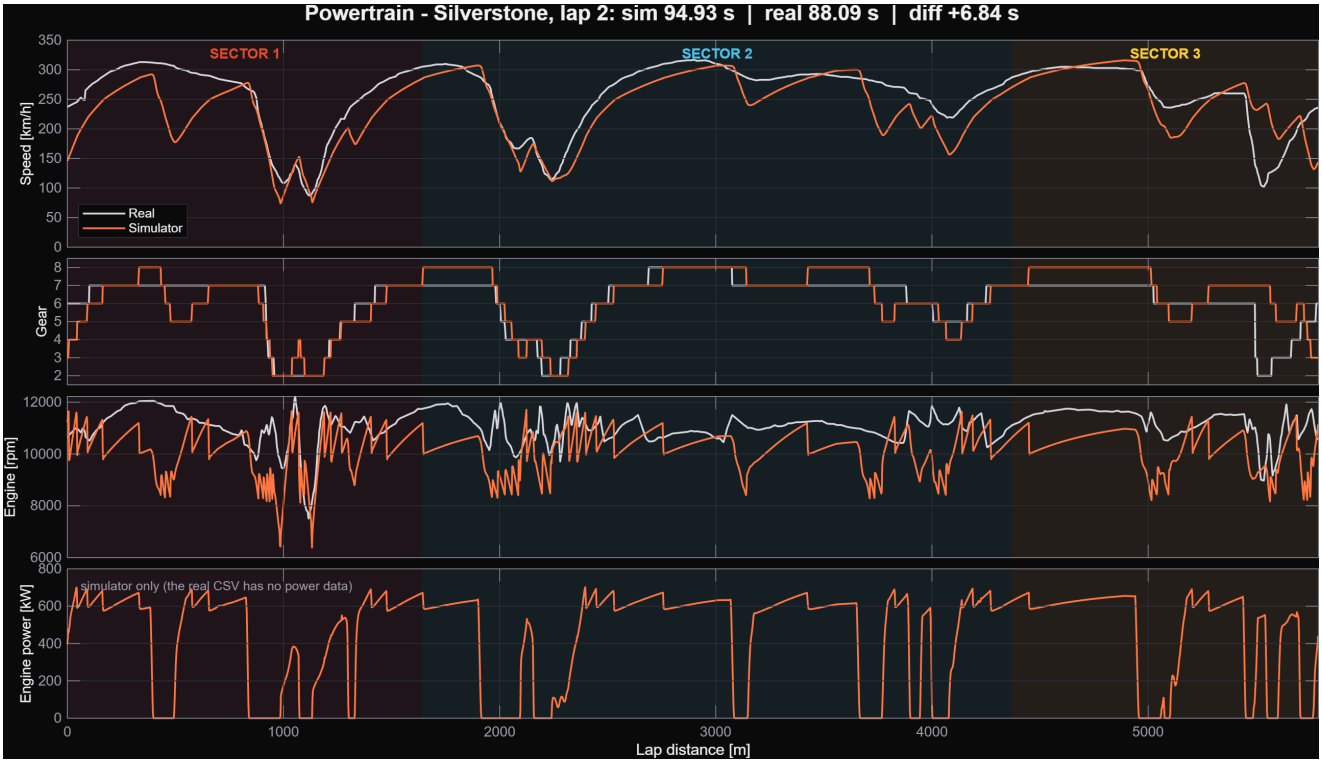


Figure 15 - Power train: speed, gear, engine rpm and engine power.

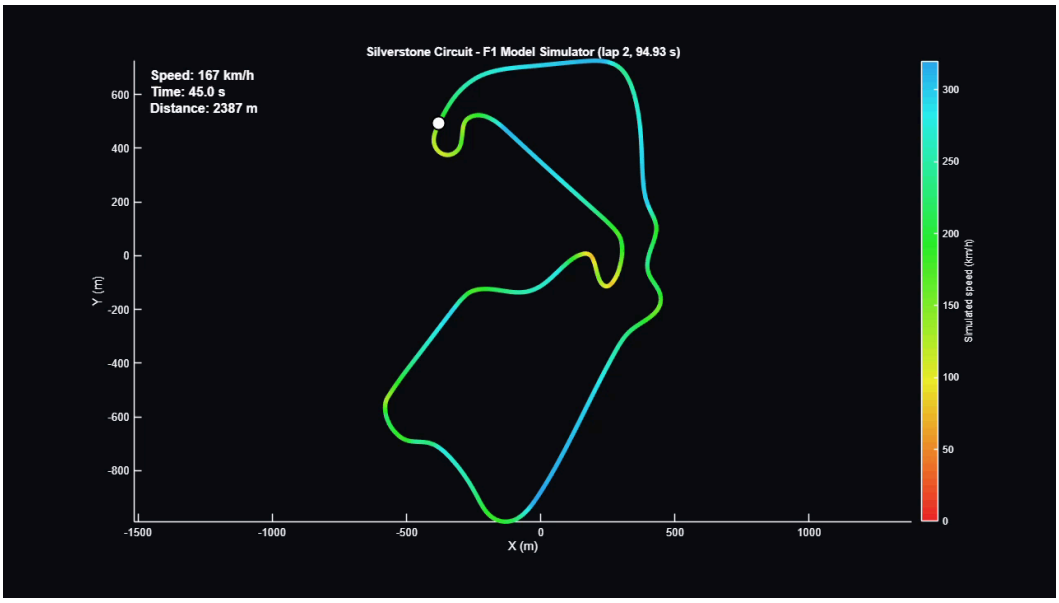


Figure 16 - Simulated lap colored by simulated speed (frame of the animation `docs/img/f1_lap.mp4`).

7.2 Findings

- **Active aerodynamics explain the top speed, not the lap time.** With a single aero configuration the car reached 302 km/h against 317 real. With the straight mode it reaches 316 km/h. The lap time moved from 95.28 s to 94.91 s; applying the mode to the car but not to the speed profile gave 94.56 s, and keeping the two coherent cost 0.35 s (cause not investigated). The simulated 90th-percentile acceleration is still below the real one: 4.4-4.8 m/s² against 5.8 at 70-80 m/s, and 1.4-2.3 against 2.9 at 80-90 m/s. Closing that with drag alone would need a CdA of about 0.6, which is implausible, so the rest is left unexplained (candidates: electric deployment, slipstream, slopes).
- **Aerodynamic level.** Moving from ClA 2.7 to 4.8 m² (inside the published range) reduced the gap from +14.9 s to +6.6 s.
- **Power.** The engine map gave about 1 MW at first; scaling it to the regulation 750 kW with a single gear made the lap 10 s slower, because one ratio keeps the engine far from its power peak for most of the lap. With an 8-speed gearbox the lap is 95.28 s with realistic power.
- **The gap is local, not a uniform lack of grip.** Speed differences are concentrated in a few corners. Comparing the curvature of the planned line with the real trajectory: near 500 m the planned corner is tighter (76 m against about 120 m; the simulator slows to 180 km/h where the real car takes it at about 300), and near 5500 m it is much more open (79 m against about 32 m; about 185 km/h against 105). Near 4050 m the radius (63 m against 81 m) explains only part of the difference; the rest looks like grip (the real car sustains about 4.7 g at 219 km/h against about 4.1 g available in the model).
- **Some gains are artifacts.** The −1.46 s in sector 3 is more likely a consequence of that geometry in slow corners than a merit of the model.
- **A hypothesis that did not hold.** "The simulator brakes too little" was tested and discarded: peak braking deceleration is similar in sectors 1 and 2, and the largest difference is in sector 3.

Computed (not simulated) with the curvature of the real trajectory, the profile's corner limit would be 304 / 197 / 113 km/h at the three points, against 299 / 219 / 102 measured. The real path is nevertheless not used as the reference line: the project builds its own geometry from public data, and

using the real path would turn part of the validation into an input. A switch exists for diagnosis (`silverstone_raceline_real.csv`) but has not been run in the simulator.

8. Limitations and known defects

Modelling level

- **L-1** Rigid body: no suspension, roll, pitch or ride height.
- **L-2** Lumped aerodynamics: no aero balance (`Cpm = 0`), no ground effect; the straight mode changes drag only.
- **L-3** Lateral grip is scaled linearly with load; there is no load sensitivity of the tire.
- **L-4** Most tire and chassis parameters are estimates (section 5).

Left out on purpose

- **No ERS.** The 2026 car deploys at most 350 kW from the electric motor and about 4 MJ per lap, about 11 s of full deployment in an 88 s lap, while the model already supplies about 700 kW continuously, which is more average power than the real car. A realistic ERS would most probably slow the lap, so adding it would complicate the model and worsen the headline number without improving fidelity. A starting point (`f1_ers.m`) is kept, unconnected.
- **No wet track, tire compounds or wear.** A switch with a single grip factor (0.82, calibrated to the crossover from slicks to intermediates at 110-112 % of the dry lap) was designed but needs about six edits in the model; it was not integrated. Compounds and wear need grip to be a signal.

Known defects

- Gear hunting in slow corners (saw-tooth engine speed) and torque spikes at each shift (`tau_s` = 0.01 s); no effect on lap time.
- Two isolated wheel-slip peaks per lap (slip ratio about 0.18 rear, 0.06 front), not investigated.
- The simulator accelerates 0.5-1 m/s² below its own computed table above 70 m/s; cause not identified.
- Geometry mismatches near 500 m, 4050 m and 5500 m (section 7.2).
- The driver does not search for its own grip limit: the grip level `mu_v` is set by hand (decision D-15).
- No suspension: load transfer is instantaneous, so the inner wheels unload fully in some corners (about 1100 m and 2100 m).
- Sector limits are approximate and real sector times are estimates.

9. Development history

Version	Lap	Gap	Main change
v1	126.58 s	+38.5 s	First validated lap, centerline following, single gear
v2	109.29 s	+21.2 s	Minimum-curvature race line; regression test; comparison aligned by position
v3	102.96 s	+14.9 s	Friction circle in the torque and brake limiters
v4	94.71 s	+6.6 s	ClA 4.8 m^2; wheel FZMAX 15 000 N; limited-slip differential
v5	95.28 s	+7.2 s	Engine at about 750 kW; 8-speed gearbox; computed acceleration table
v6	94.91 s	+6.8 s	Active aero straight mode through the external force port
v7	94.93 s	+6.8 s	Single source of parameters, sensitivity sweep, internal-states view; documentation in English

Every step was one change at a time, followed by the regression test. The decision log records the reasoning behind the main choices (for example why the ERS and the real GPS path were left out). Between v3 and v4, aero and wheel load limit were changed together, so their separate effects are not isolated; v5 is slightly slower than v4 on purpose, because it trades a 1 MW engine with one gear for realistic power and transmission.

10. Roadmap

1. Driver limit search: global, then per corner, then online grip estimation (paused; D-15).
2. ERS with a per-lap energy budget, starting from `f1_ers.m` (it would first need a lower continuous power).
3. Wet track and tire compounds, then wear and temperature.
4. Load sensitivity of grip and aero balance, to address limitations L-2 and L-3.
5. Shift logic with a minimum time between shifts; reduction of the geometry differences listed in section 7.2.
6. Race-level features: fuel consumption, degradation, strategy.

11. Reproducing the results

1. Open the project `F1.prj` in MATLAB R2026a with Simulink and Vehicle Dynamics Blockset.
2. Run `f1_vehicle_params`: it loads the vehicle, circuit, race line, speed profile and finish line.
3. Simulate `F1_Simulator` for 260 s.
4. Run `f1_regression_test` (PASS/FAIL), then use the *Analysis tools* buttons of the model (telemetry, sector map, animation).

References

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9. OpenStreetMap contributors, circuit geometry, ODbL. openstreetmap.org/copyright

Aerodynamic and tire values from secondary sources are used only as orders of magnitude; the final model values are assumptions and are declared as such in the parameter register.

Appendix A. License and attribution

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Appendix B. Document set

File	Content
`README.md`	Overview, results, quick start, limitations
`docs/architecture.md`	Block diagrams and signal flow
`docs/parameter_register.md / .csv`	Every parameter, value, where it is used and its status
`docs/requirements_traceability.md`	Requirements, verification methods, traceability to the regression test
`docs/decision_log.md`	Engineering decisions and their reasons
`docs/CHANGELOG.md`	Versions v1-v7